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WORK EXPERIENCE

Commision on Higher Education CALABARZON (CHED Region IV).

Full Stack Developer Intern

Onsite, May 2026 – Present

- Developed a web-based **Graduate Tracer System** designed for Higher Education Institutions (HEIs) **across the CALABARZON region**, with potential for future nationwide implementation.
- Built the system from scratch using **PHP Laravel** for the backend and **React TypeScript** for the frontend.
- Designed and structured the **system architecture**, **database schema**, and **ERD** based on the program flow, operational requirements, and recommendations provided by supervisors.
- In collaboration with supervisors, I implemented core web application features, interfaces, and workflows to support graduate tracking, institutional data management, and reporting processes.

JLabs Innovatech Inc.

Full Stack Developer Intern

Remote, March 2026 – April 2026

- Contributed to the development of a cross-platform mobile application for **iOS** and **Android** using **React Native** and **Supabase**
 - Collaborated with the team to **refine and restructure an AI-generated codebase**, improving code quality, readability, and maintainability, while integrating AI features using **Anthropic** APIs
 - Implemented major frontend UI revisions to enhance usability and overall user experience
 - Optimized search functionality, improving performance and responsiveness for a user base of **~1,000** users
 - Collaborated with cross-functional teammates to ensure smooth frontend–backend integration
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PROJECTS

Barako Sense: Developing a Sensory Lexicon for Kapeng Barako || Thesis

- Led the **AI engineering and model development** for a hybrid research project aimed at creating the **first sensory lexicon for Kapeng Barako (Coffea Liberica)** in Batangas.
- The study combined **CNN-based tree identification** with **traditional coffee cupping** and sensory evaluation.
- Collected and processed **4,000+ images** of bark, leaves, and cherries from **5 DNA-tested Liberica trees**, in collaboration with **De La Salle University Manila**.
- Designed and trained a **MobileNetV2 model** with a **multi-input, single-classification pipeline**, achieving **95% accuracy** in distinguishing individual Liberica trees.
- Contributed to the creation of a **sensory and physical traits dataset** with **100+ cupping entries**, supporting future agricultural and sensory research on Batangas Barako coffee.

Predicting Enrollment Probability Using KNN and Random Forest || NU-Lipa Project

- I led the **data science and model optimization** efforts, focusing on **feature engineering**, **model tuning**, and **performance evaluation**.
- Improved the existing predictive model's accuracy from **83% to 90.67%**, achieving a **7% performance increase**.
- Utilized **K-Nearest Neighbors (KNN)** and **Random Forest** algorithms to predict enrollment likelihood based on **20,000+ official student application records** from NU-Lipa.
- Enhanced data preprocessing and variable selection for better model interpretability and decision support for admissions officers.

NPK Deficiency Detector in Bitter Gourd Leaves || Software Engineering Project

- Developed a **Deep Learning CNN model using ResNet50** to classify bitter gourd leaves as *healthy* or *deficient in nitrogen, phosphorus, or potassium*.
- As the lead **AI engineer and data scientist**, I handled dataset preparation, model design, and performance tuning.

- Achieved **95% accuracy**, with an **F1 score of 96%**, **Precision of 96%**, and **Recall of 96%**.
- Trained and tested on **2,500+ verified images**, sourced online and through local data validated by the **Department of Agriculture IV-A**.
- Focused on improving model generalization and minimizing class imbalance for real-world agricultural applications.
- Featured on **International Conference on Machine Learning & Computer Research ICMCR (2026)** in Tokyo Japan.

Legal Aid at Your Access (LAYA) || iNUvators, APPCON 2023 Submission

- Developed a bilingual conversational AI for Philippine law using **Microsoft Azure's Conversational Language Understanding (CLU)**.
- I wrote the python script for web scraping over **3,000+** legal documents from sources such as **Lawphil.net** and various local law firms, covering Republic Acts, Constitutional Acts, and other public legal texts.
- I also contributed to **data annotation** and **model context optimization** within Azure to improve entity recognition and response accuracy.
- The model achieved an **F1 score of 84.43%**, **Precision of 85.22%**, and **Recall of 83.65%** across **3,000 training** and **600** testing utterances.
- Focused on data preprocessing, multilingual intent handling, and legal-domain entity extraction for AI-assisted legal information access.

SKILLS

Langguages: JavaScript, TypeScript, Python, SQL, Java, PHP

Web Frameworks: React, Django, Node.js, Flask, FastAPI

Data Science & ML: Tensorflow/PyTorch, Sci-Kit Learn, Numpy, Pandas, Hugging Face Transformers

Visualization & Tools: RapidMiner, Gradio, Weights and Biases, PowerBI, Excel/Google Sheets, Matplotlib

Operating System: Windows, Linux

EDUCATION

National University

Lipa City, Batangas

BS in Computer Science – Specialization in Machine Learning, **GWA: 3.51/4.00**

2022 - 2026

Relevant Coursework: Systems Programming, Data Structures and Algorithms, Adv anced Machine Learning, Deep Learning, Software Engineering, Data Science

iNUvators member, **8-time** Deans Lister, TOEIC Exam Score: **925**

ACTIVITIES AND PARTICIPATIONS

CHED LAKAS

National University Lipa, October 25, 2023

Certificate of Participation Holder

A conference for computing educators, researchers, and professionals in the Philippines and nearby countries.

APPCON 2023

National University Lipa, Oct. 2023 – April 2024

Represented NU-Lipa as part of 'iNUvators' Nationwide

Submitted "LAYA", an AI application in terms of the theme "A web or mobile application to address social issues in the Philippines"

24th Philippine Computing Science Congress

De La Salle Laguna, May 9-11, 2024

Base Build Calabarzon

National University Lipa, September 10, 2025

A seminar for blockchain development, within a day of hands-on, learning, collaboration and on chain creation.

- Achieved 3rd place in the hackathon with **"Arcade Hub,"** a project developed and pitched within the event

CERTIFICATIONS

IC3 Digital Literacy Global Standard Six • Information Technology Specialist: Python • Machine Learning w/ Python

• **AI Agents Fundamentals • LLM Course Unit 3: Fine-Tuning Language Models**

